

How do we create subgroups of treatment and control males with more similar distributions of age than is seen overall, in fact, so similar that we could believe that the data arose from a randomized block experiment? Cochran’s example used subclassification. First, the smokers are divided at the overall median into young smokers and old smokers—two subclasses, and then divided into young, middle aged, and old smokers, each of these three subclasses being equal size, and so forth. Finally, nine subclasses are used. The age distributions within each of the nine subclasses are very similar for the treatment condition and the control condition, just as if the men had been randomly assigned within the age subclasses to treatment and control, because there is such a narrow range of ages within each of the nine subclasses. And of great importance, there do exist both treatment and control males in each of nine subclasses.

The design phase can be considered complete for our simple illustrative example. Our underlying hypothetical randomized experiment that led to the observed dataset is a randomized block experiment with nine blocks defined by age, where the probability of being assigned to the treatment condition (cigarette smoking) rather than the control condition (cigar/pipe smoking) decreases with age. We are now allowed to look at the outcome data within each subclass and compare treatment and control death rates. We find that, averaging over the nine blocks (subclasses), the death rates are about 50% greater for the cigarette smokers than the cigar and pipe smokers. Incidentally, the full data set with no subclassification leads to nearly the opposite conclusion; see [Cochran \(1968\)](#) or [Rubin \(1997\)](#) for details.

But what would have happened if we decided that we wanted to subclassify also on education, socio-economic status, and income, each covariate using, let’s say, five levels [a minimum number implicitly recommended in [Cochran \(1968\)](#)]? There would be four key covariates, each with five levels, yielding a total of 625 subclasses. And many observational studies have many more than four key covariates that are known to be used for making treatment decisions. For example, with 20 such covariates, even if each is dichotomous, there are 2^{20} subclasses—greater than a million, and as a result, many subclasses would probably have only one unit, either a treated or control, with no treatment-control comparison possible. How should we design this step of observational studies in such more realistic situations?

4.2. Propensity score methodology. [Rosenbaum and Rubin \(1983\)](#) proposed a class of methods to try to achieve balance in observational studies when there are many key covariates present. In recent years there has been an explosion of work on and interest in these methods; the Introduction in [Rubin \(2006\)](#) offers some references. Sadly, many of the articles that use propensity score methods do not use them correctly to help design observational studies according to the guidelines in Section 3, which are motivated

by the theoretical perspective of Section 2 and illustrated in the trivial one-covariate example of Section 4.1. Rather, these inappropriate applications, for example, use the outcome data to help choose propensity score models, and use the propensity score only as a predictor in a regression model with the outcome, Y_{obs} , as the dependent variable.

The propensity score is the observational study analogue of complete randomization in a randomized experiment in the sense that its use is not intended to increase precision but only to eliminate systematic biases in treatment-control comparisons. In some cases, however, its use can increase precision; for the reason, see [Rubin and Thomas \(1992\)](#). As we have seen in earlier sections, it is formally defined as the probability of a unit receiving the treatment condition, rather than the control condition, as a function of observed covariates, including indicator variables for the individual decision makers and associated interactions, if needed. The propensity score is rarely known in an observational study, and therefore must be estimated, typically using a model such as logistic regression, but this choice, although common, is by no means mandatory or even ideal in many circumstances. The critical aspect of the propensity score is that it models the reasons for treatment versus control assignment at the level of the decision maker. For instance, in the context of the expanded tobacco example of Section 4.1, it could model the choice of a male smoker to smoke cigarettes versus cigars or pipes as a function of age, income, education, SES, etc. Once estimated, the linear version of it (e.g., the beta times X in the logistic regression) can be treated as the only covariate, just like age in the example of Section 4.1, and it is used to match or subclassify the treatment and control units.

But we are not done yet. We have to check that balance on all covariates has been achieved. If the propensity score is correctly estimated and there is balance on it, then [Rosenbaum and Rubin \(1983\)](#) showed that balance is achieved on all observed covariates. The achieved balance within matched pairs or subclasses must be assessed and documented before the design phase is finished. With only one covariate, balance on that covariate is easily achieved (if it can be achieved) by using narrow enough subclasses (or bins) of the covariate. With many covariates, the assessment and re-estimation of propensity score to achieve balance can be tricky, and good guidance for doing this is still being developed. When selecting matched pairs, using both the propensity score and some prognostically important functions of key covariates can often result in increased precision of estimation [see [Rubin \(1979b\)](#), [Rosenbaum and Rubin \(1985\)](#), [Rubin and Thomas \(2000\)](#)].

Here we illustrate these various ideas in the context of some real examples. The next example concerns the relative success of two treatments for breast cancer, and illustrates not only the process of selecting the key background variables for use in the propensity score estimation, but also illustrates that

careful observational studies can (not necessarily will) reach the same general conclusions as expensive randomized experiments. The second example is from a large marketing study and displays the kind of balance that can be achieved following propensity score subclassification, as well as the fact that some units can be unmatchable. The last example, in Section 5, uses a data set on large volume versus small volume hospitals to emphasize that one observational data set can be used to support two (or more) differing templates for the underlying randomized study of a particular question, and one template may be considered far better than the other.

4.3. *GAO study of treatments for breast cancer.* The following example appeared in a Government Accounting Office (GAO) publication that was summarized in Rubin (1997). In the 1960s mastectomy was the standard treatment for many forms of breast cancer, but there was growing interest in the possibility that for a class of less severe situations (e.g., small tumors, node negative) a more limited surgery, which just removed the tumor, might be just as successful as the more radical and disfiguring operation.

Several large and expensive randomized trials were done for this category of women with less severe cancer, and the results of these trials are summarized in Table 1. As can be seen there, these studies suggest that for this class of women who are willing to participate in a randomized experiment, and for these cancer treating centers and their doctors, who are also willing

TABLE 1
Estimated 5-year survival rates for node-negative patients in six randomized clinical trials

Study	Women	Women	Estimated survival rate for women	Estimated survival rate for women	Estimated causal effect
	Breast conservation (BC)	Mastectomy (Mas)	BC	Mas	BC–Mas
Study	<i>n</i>	<i>n</i>	%	%	%
U.S.–NCI†	74	67	93.9	94.7	−0.8
Milanese†	257	263	93.5	93.0	0.5
French†	59	62	94.9	96.2	−1.3
Danish‡	289	288	87.4	85.9	1.5
EORTC‡	238	237	89.0	90.0	−1.0
U.S.–NSABP‡	330	309	89.0	88.0	1.0

† Single-center trial; ‡ Multicenter trial.

Reference: Rubin, D. B. Estimated causal effects from large datasets using propensity scores. *Annals of Internal Medicine* (1997); 127, 8(II):757–763.

to participate, the five-year survival rate appears to be very similar in the two randomized treatment conditions. There is, however, an indication from Table 1 that the survival is better overall in the trials conducted in single centers (the top three rows) than in the multi-center trials (the bottom three rows), possibly because of more specialized care, including after care.

The reason this last comment is relevant is that based on these results, the U.S. National Cancer Institute felt that for this category of women, the recommendation for general practice should be to have breast conserving operations rather than the radical version. The GAO was concerned that this advice based on these randomized trials may not be wise for general practice, where the surgeons involved may not be as skilled, after care may be lower quality, the women themselves may be less research-oriented and therefore less medically astute about their own care, and so forth, than in the randomized trials. It was not possible to initiate a new randomized trial in the general population of women and doctors who may not want to be randomized; even if it were, the funding, planning, implementing, etc., would take too long and results concerning five-year survival would be first available a decade in the future.

Consequently, the GAO implemented an observational study using the SEER (Surveillance, Epidemiology, End Results) data base, which has relatively complete information on all cancer cases in certain catchment areas. Importantly, it had detailed information on the kind and diagnosed severity of the cancer, so that they could use the same exclusion criteria as the randomized experiments, and it had the kind of treatment used by the surgical team; also, it had survival information, which of course was the key outcome variable. Moreover, it had many covariates. And it had about five thousand breast cancer cases of the type studied in the six randomized experiments during the relevant period, which was considered a large enough sample to proceed.

So far so good. The outcome data were stripped from the files, and the design phase proceeded. The following description is from an over 15 year old memory, and no doubt is somewhat distorted by my current attitudes, but is largely accurate, I believe. The GAO checked with a variety of physicians about who the decision makers were for choice of surgery for this category of women. The replies were that they were usually joint choices made by the surgeon and woman, sometimes with input from the husband or other family members or friends. Some of the key covariates were obvious, such as the size of the tumor and the woman's age and marital status. Others were less obvious, such as urbanization, region of the country, year, race, and various interactions (e.g., age by marital status). In any case, a list of approximately twenty key covariates was assembled, and it turned out that all had been collected in SEER. More good news. Then the consistency of the decision makers' rules across the dataset was considered, although